

# Wealth and the Structure of National Power

How does GDP per capita relate to different dimensions of national power?

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# Overview

National power is central to debates in international politics, but it is inherently multidimensional. This project examines whether wealth translates uniformly across different components (economic, military, and technological) of state power.

## **Central Findings**

- Wealth is positively associated with all dimensions of national power, but not equally across them.
- The strength of this relationship varies across economic, military, and technological power.
- Some countries deviate from the general pattern.

# The Data: Global Power and Economic Indicators

## Dataset: Global Power and Economic Indicators

- This dataset has 250 observations and combines publicly available data from the World Bank Open Data API and the REST Countries API.
  - The dataset was inspected to ensure that all observations correspond to individual countries, and no regional or aggregate entries were included.
- The dataset includes indicators of economic, technological, and military capacity, as well as a composite measure of national power for countries.
- The core variables for this project are:
  - country
  - gdp\_per\_capita
  - economic\_power ( $\text{GDP} \times \text{GDP growth}$ )
  - tech\_power ( $\text{Internet usage} \times \text{High-tech exports}$ )
  - military\_power ( $\text{Military spending}$ )
  - power\_index (composite national power based on all power components)
  - power\_rank

# Inconsistencies in GDP per Capita and National Power

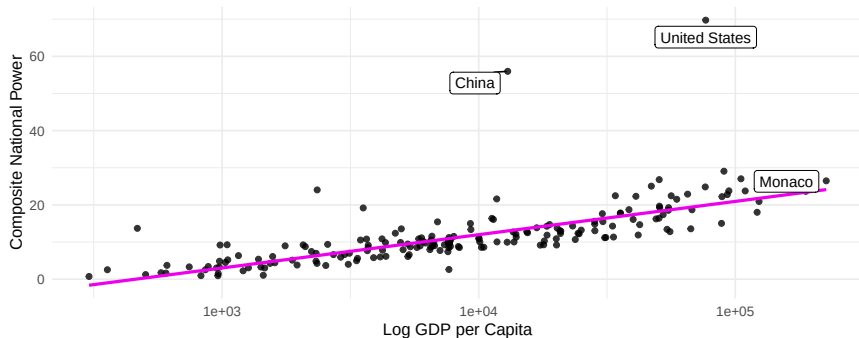


Figure 1: GDP per Capita and Composite National Power

- As demonstrated by the Monaco and China outliers in Figure 1, high GDP per capita does not reliably translate into high national power.

# Strength of the Relationships: Regression Results

Table 1: Relationship Between GDP per Capita and Power Dimensions

| Power Component | Beta Coefficient | P Value |
|-----------------|------------------|---------|
| Power Index     | 2.000000e-04     | 0.0000  |
| Economic Power  | 3.259892e+07     | 0.0144  |
| Tech Power      | 1.070000e-02     | 0.0000  |
| Military Power  | 3.120499e+05     | 0.0159  |

- The estimated coefficients are larger for economic and military power than for technological power, indicating that wealth is more strongly associated with economic and military dimensions of power than with technological power.

# Wealth and the Dimensions of National Power

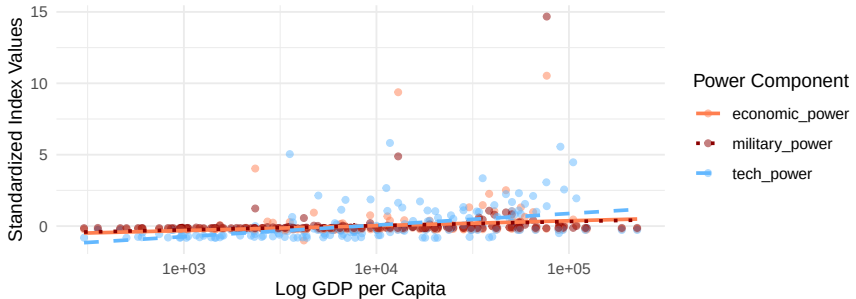


Figure 2: GDP per Capita and Different Dimensions of National Power

- Figure 2 shows that GDP per capita aligns more closely with economic and military outcomes than with technological outcomes, where the relationship is weaker and more dispersed.

# Conclusion

- Wealth is consistently associated with higher levels of national power, but the strength of this relationship varies across different dimensions.
- Table 1 and Figure 2 suggest that wealth more reliably translates into economic and military capacity than into technological capacity, indicating that GDP per capita operates more as a general enabling resource than a uniform driver of national power.
- Outliers such as Monaco and China highlight that state size and strategic priorities can condition how effectively wealth translates into power.

**Overall, wealth is a necessary but insufficient foundation for national power; it enables, but does not guarantee, a state's ability to project influence across economic, military, and technological domains.**